Design and Simulation of a Circular Microstrip Patch Antenna for RFID Applications at 2.45 GHz

Simulated in CST Studio Suite

V SHRUTHI SHRI

***Abstract* —** This work presents the design and simulation of an inset-fed circular microstrip patch antenna for radio frequency identification (RFID) in the 2.45 GHz ISM band. The antenna is built on a low-cost FR4 substrate (εr = 4.3, h = 1.6 mm) and simulated in CST Studio Suite using the transient solver from 1.5 to 3.5 GHz. The simulated antenna resonates at about 2.42 GHz with a minimum S11 of about –15.5 dB, and its –10 dB impedance bandwidth of roughly 2.38–2.47 GHz covers the 2.45 GHz operating frequency. At 2.45 GHz the antenna shows a directivity of 6.0 dBi, a gain of 2.27 dBi and a wide broadside beam with 3 dB beamwidths of 107.2° and 90.9° in the two principal planes. The compact planar design is a practical candidate for low-cost 2.45 GHz RFID readers and tags.

***Keywords* —** *RFID, circular microstrip patch antenna, 2.45 GHz ISM band, FR4, CST Studio Suite, return loss, radiation pattern.*

I. INTRODUCTION

Radio frequency identification (RFID) enables fast, contactless identification of objects and is now used well beyond inventory tracking, in areas such as smart agriculture [1], wearable biomedical monitoring [2], environmental sensing [3] and structural health monitoring [4]. An RFID system consists of a reader, a tag and the antennas that link them, and the antenna largely determines the system's read range, bandwidth and reliability [5], [6].

Microstrip patch antennas are widely used in RFID because they are compact, lightweight, low-cost and easy to integrate with planar circuits [7]. Circular patches in particular offer a smaller footprint than rectangular patches of the same resonant frequency. Recent RFID antenna research covers a wide range of designs, including printed UHF tags for dielectric and metallic objects [8], flexible inkjet-printed tags [9], textile chipless tags for wearable use [10], dual-band circularly polarized reader antennas [11], integrated NFC/UHF/2.45 GHz reader antennas [12] and compact active-tag antennas [13].

In this work, an inset-fed circular microstrip patch antenna for the 2.45 GHz ISM band is designed on an FR4 substrate and simulated in CST Studio Suite. Its return loss, VSWR and radiation characteristics are analysed and compared with published RFID antenna designs.

II. ANTENNA DESIGN

The radius of a circular microstrip patch for a resonant frequency fr is estimated from the cavity model [7]. An intermediate quantity F is first calculated as

$$F = \frac{8.791 \times 10^9}{f_r \sqrt{\varepsilon_r}} \tag{1}$$

with fr in hertz and F in centimetres. The physical radius a, which accounts for fringing fields, is then

$$a = \frac{F}{\left\{1 + \frac{2h}{\pi \varepsilon_r F}\left[\ln\left(\frac{\pi F}{2h}\right) + 1.7726\right]\right\}^{1/2}} \tag{2}$$

where h is the substrate thickness in centimetres and εr is the relative permittivity. For fr = 2.45 GHz, εr = 4.3 and h = 0.16 cm, (1) gives F ≈ 1.73 cm and (2) gives a ≈ 16.8 mm. The simulated model uses a radius of 17.35 mm. The patch is fed by a 50 Ω microstrip line with a 10 mm inset, which moves the feed point inside the patch to improve impedance matching. The final design parameters are listed in Table I and the geometry is shown in Fig. 1.

TABLE I
DESIGN PARAMETERS

Parameter	Value
Design frequency (fr)	2.45 GHz
Substrate	FR4 (lossy)
Relative permittivity (εr)	4.3
Loss tangent (tan δ)	0.02
Substrate thickness (h)	1.6 mm
Patch radius, calculated from (2)	16.78 mm
Patch radius, simulated model (a)	17.35 mm
Ground plane (gp)	1.5 mm
Feed line width (wf)	3.13 mm
Feed inset (fi)	10 mm
Feed line length (lf)	14.75 mm
Port impedance	50 Ω

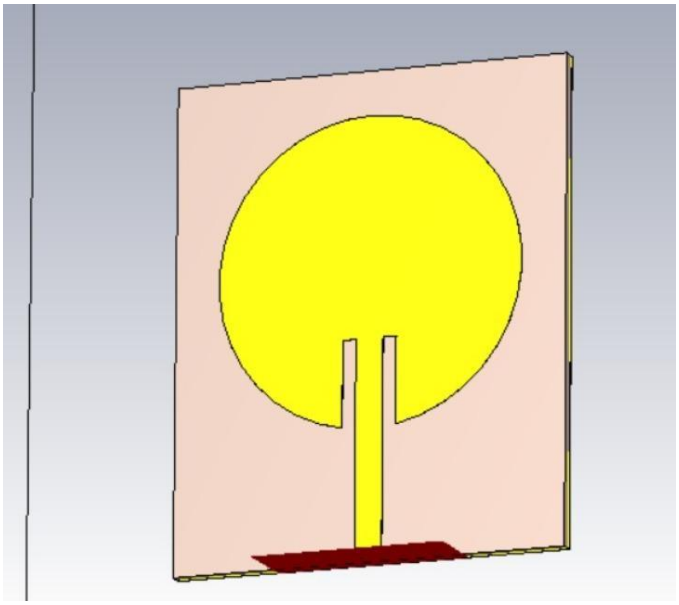


Fig. 1. Geometry of the proposed circular patch antenna in CST Studio Suite.

III. SIMULATION SETUP

The antenna was modelled and simulated in CST Studio Suite with the following configuration:

- Solver: transient (time-domain), for broadband analysis
- Frequency range: 1.5 – 3.5 GHz
- Substrate: FR4 (lossy), $\varepsilon r = 4.3$, $\tan \delta = 0.02$, $h = 1.6$ mm
- Feed: 50 Ω microstrip line
- Far-field monitor: 2.45 GHz

IV. RESULTS AND DISCUSSION

A. Return Loss (S11)

Fig. 2 shows the simulated reflection coefficient S11. The curve has a single, well-defined resonance near 2.42 GHz, where S11 reaches a minimum of about –15.5 dB. Since an S11 below –10 dB is the usual benchmark for acceptable matching, the antenna's –10 dB impedance bandwidth extends from roughly 2.38 to 2.47 GHz, about 90 MHz or 3.7 % of the centre frequency. This band includes the 2.45 GHz design frequency and most of the 2.40–2.4835 GHz ISM band.

The resonance sits about 30 MHz below the 2.45 GHz target. This is consistent with the simulated radius (17.35 mm) being slightly larger than the value calculated from (2) (16.8 mm), since a larger patch resonates at a lower frequency. Reducing the radius slightly would centre the resonance at 2.45 GHz.

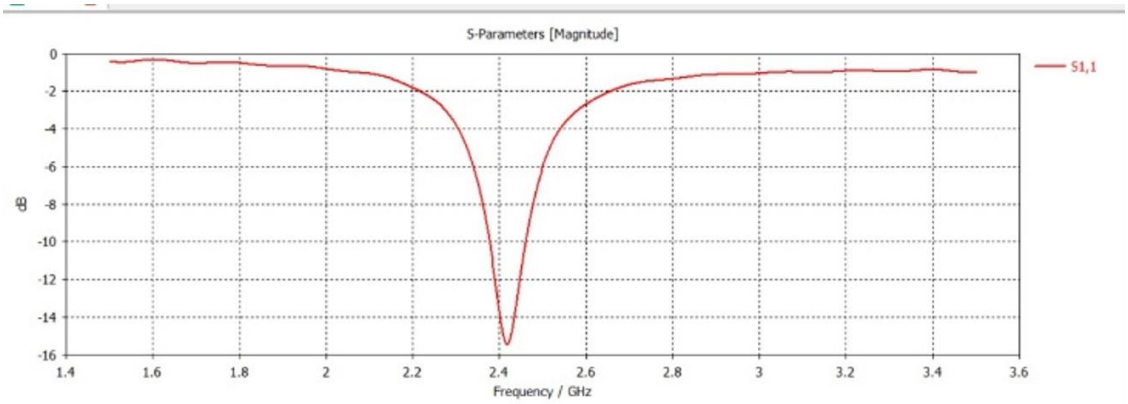


Fig. 2. Simulated S11 of the proposed antenna.

B. Voltage Standing Wave Ratio (VSWR)

The simulated VSWR is shown in Fig. 3. It reaches its minimum at the resonance, where an S11 of –15.5 dB corresponds to a VSWR of about 1.4. A VSWR below 2 is equivalent to an S11 below about –9.5 dB, so the VSWR stays below 2 across the same band identified in Fig. 2. Outside this band the mismatch rises sharply, as expected for a narrowband patch antenna.

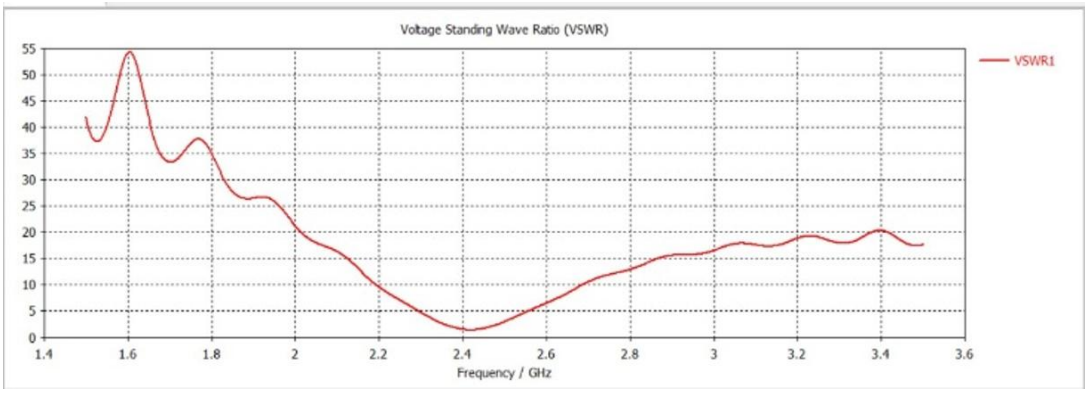


Fig. 3. Simulated VSWR of the proposed antenna.

C. Radiation Characteristics

Figs. 4 and 5 show the simulated 3D directivity and gain patterns at 2.45 GHz. The antenna has a maximum directivity of 6.0 dBi and a gain of 2.27 dBi. The 3.73 dB difference between the two equals the simulated radiation efficiency of –3.73 dB (about 42 %); the total efficiency, which also includes mismatch loss, is –4.05 dB (about 39 %). The relatively low efficiency is largely attributable to the lossy FR4 substrate (tan δ = 0.02).

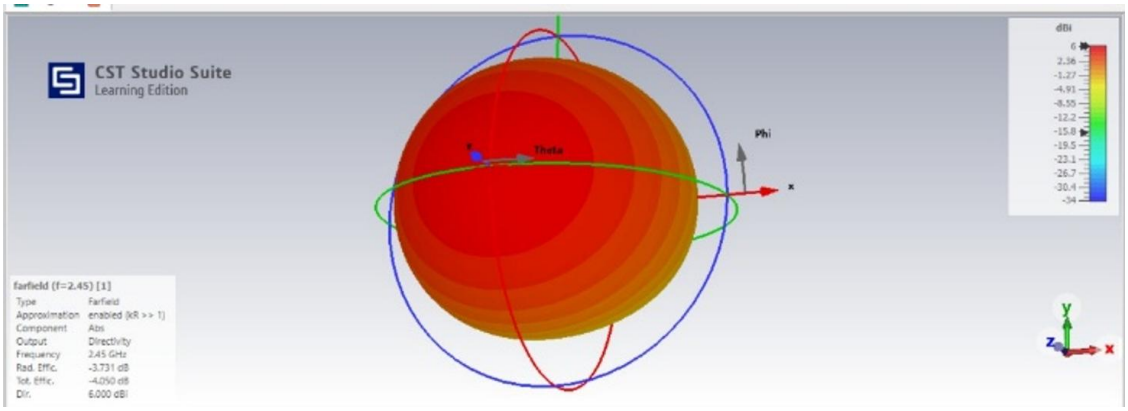


Fig. 4. Simulated 3D directivity pattern at 2.45 GHz (directivity = 6.0 dBi).

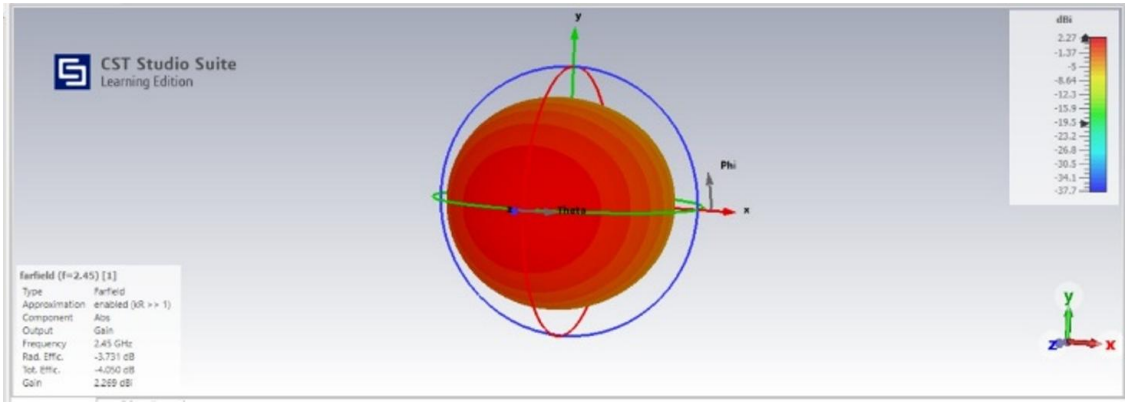


Fig. 5. Simulated 3D gain pattern at 2.45 GHz (gain = 2.27 dBi).

The 2D pattern cuts in Figs. 6–8 confirm a broadside radiation pattern, with the main lobe directed perpendicular to the patch (θ = 0°). The 3 dB beamwidth is 107.2° in the φ = 0° plane and 90.9° in the φ = 90° plane, and the side lobe level in the φ = 90° plane is –13.4 dB. In the θ = 90° plane, which lies in the plane of the patch, the pattern has a two-lobed shape with a

maximum of –1.04 dBi. The antenna is therefore not omnidirectional, but its wide main beam gives broad angular coverage in front of the antenna, which suits RFID reader applications.

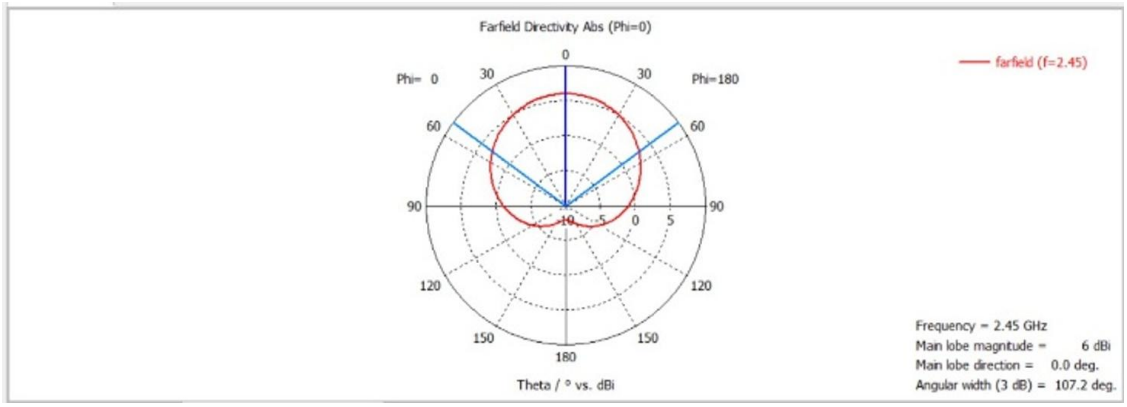


Fig. 6. Simulated directivity pattern at 2.45 GHz, φ = 0° plane.

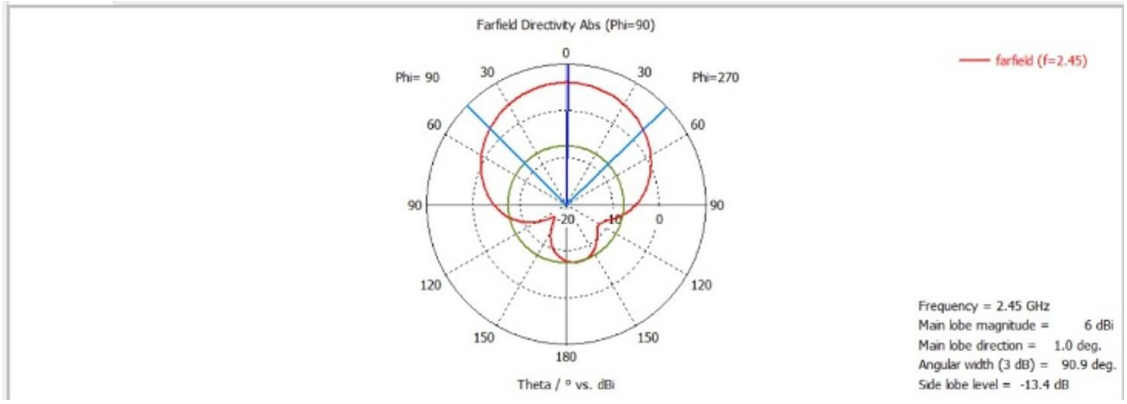


Fig. 7. Simulated directivity pattern at 2.45 GHz, φ = 90° plane.

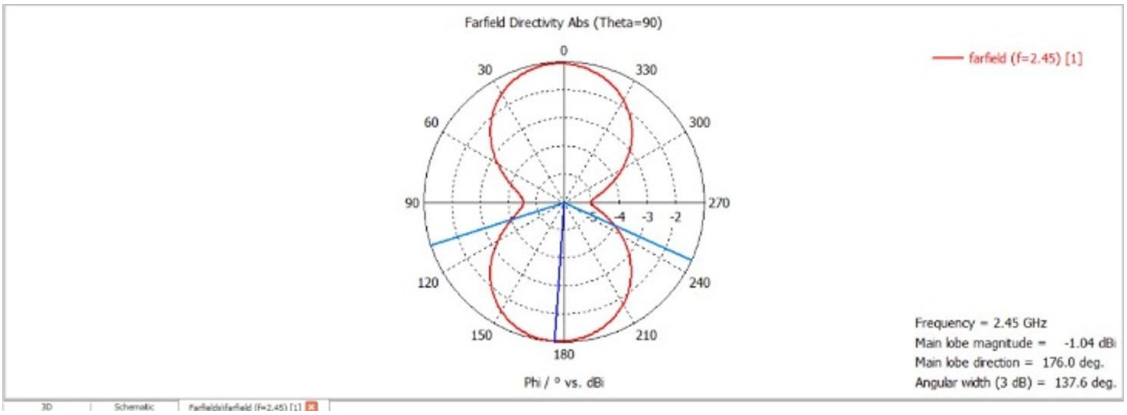


Fig. 8. Simulated directivity pattern at 2.45 GHz, θ = 90° plane.

D. Summary of Results

TABLE II
SUMMARY OF SIMULATED PERFORMANCE

Parameter	Simulated value
Resonant frequency	≈ 2.42 GHz
Minimum S11	≈ –15.5 dB
–10 dB impedance bandwidth	≈ 2.38 – 2.47 GHz (≈ 90 MHz)

Parameter	Simulated value
Minimum VSWR	≈ 1.4
Directivity at 2.45 GHz	6.0 dBi
Gain at 2.45 GHz	2.27 dBi
Radiation efficiency	–3.73 dB (≈ 42 %)
3 dB beamwidth (φ = 0° / 90°)	107.2° / 90.9°
Side lobe level (φ = 90°)	–13.4 dB

E. Comparison with Published RFID Antennas

Table III compares the proposed antenna with published RFID antenna designs. The reported designs target different frequency bands and applications, so the comparison is qualitative. Most published tag antennas operate in the UHF band and focus on flexible or wearable materials, whereas the proposed design targets the 2.45 GHz microwave RFID band on a standard, low-cost FR4 substrate.

TABLE III
COMPARISON WITH PUBLISHED RFID ANTENNAS

Reference	Application	Frequency	Substrate / material	Reported performance
Islam et al. [9]	Flexible UHF tag	Universal UHF RFID band	Photo paper, inkjet-printed silver	Read range ≈ 4.57 m (15 ft)
Vena et al. [10]	Wearable chipless tag	3 – 6 GHz	Cotton textile, sewn conductive thread	Est. read range 1.68 m
Liu et al. [11]	Handheld reader	0.92 / 2.45 GHz	—	Dual-band, circularly polarized, directional
Romputtal and Phongcharoenpanich [12]	Integrated reader	13.56 MHz, 920–925 MHz, 2.45 GHz	—	NFC + dual-band RFID in one antenna
This work	**2.45 GHz RFID**	**2.38 – 2.47 GHz (– 10 dB)**	**FR4**	**Gain 2.27 dBi, directivity 6.0 dBi**

V. CONCLUSION

An inset-fed circular microstrip patch antenna for 2.45 GHz RFID applications was designed on an FR4 substrate and simulated in CST Studio Suite. The antenna resonates at about 2.42 GHz with a minimum S11 of about –15.5 dB, and its –10 dB impedance bandwidth of roughly 2.38–2.47 GHz covers the 2.45 GHz operating frequency with a VSWR below 2. At 2.45 GHz it provides a directivity of 6.0 dBi, a gain of 2.27 dBi and a wide broadside beam with 3 dB

beamwidths of 107.2° and 90.9°. Its compact planar geometry and low-cost substrate make it a practical candidate for 2.45 GHz RFID readers and tags.

This work is limited to simulation. Future work includes fine-tuning the patch radius to centre the resonance at 2.45 GHz, fabricating a prototype and measuring its S11 and radiation pattern with a vector network analyzer, and exploring gain and bandwidth enhancement techniques such as metasurface reflectors [14] and compact monopole-type radiators [15].

REFERENCES

[1] R. Rayhana, G. Xiao, and Z. Liu, "RFID sensing technologies for smart agriculture," *IEEE Instrum. Meas. Mag.*, vol. 24, no. 3, pp. 50–60, May 2021, doi: 10.1109/MIM.2021.9436094.

[2] S. K. Behera and N. C. Karmakar, "Wearable chipless radio-frequency identification tags for biomedical applications: A review," *IEEE Antennas Propag. Mag.*, vol. 62, no. 3, pp. 94–104, Jun. 2020, doi: 10.1109/MAP.2020.2983978.

[3] A. Vena, E. Perret, D. Kaddour, and T. Baron, "Toward a reliable chipless RFID humidity sensor tag based on silicon nanowires," *IEEE Trans. Microw. Theory Techn.*, vol. 64, no. 9, pp. 2977–2985, Sep. 2016, doi: 10.1109/TMTT.2016.2594229.

[4] W. Chompoosawat, A. Boonpoonga, P. Akkaraekthalin, L. Bannawat, and T. Lertwiriyaprapa, "Single-layer chipless RFID sensor for metal crack detection," in *Proc. Int. Elect. Eng. Congr. (iEECON)*, 2021, pp. 575–578.

[5] K. Finkenzeller, *RFID Handbook: Fundamentals and Applications in Contactless Smart Cards, Radio Frequency Identification and Near-Field Communication*, 3rd ed. Chichester, U.K.: Wiley, 2010.

[6] N. S. Khair, N. A. T. Yusof, M. H. M. Ariff, Y. A. Wahab, and B. S. Bari, "Recent advances and open challenges in RFID antenna applications," in *Enabling Industry 4.0 through Advances in Mechatronics* (Lecture Notes in Electrical Engineering, vol. 900). Singapore: Springer, 2022, pp. 507–517, doi: 10.1007/978-981-19-2095-0_43.

[7] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed. Hoboken, NJ, USA: Wiley, 2016.

[8] K. Siakavara, S. Goudos, A. Theopoulos, and J. N. Sahalos, "Passive UHF RFID tags with specific printed antennas for dielectric and metallic objects applications," *Radioengineering*, vol. 26, no. 3, pp. 735–745, Sep. 2017, doi: 10.13164/re.2017.0735.

[9] M. T. Islam, T. Alam, I. Yahya, and M. Cho, "Flexible radio-frequency identification (RFID) tag antenna for sensor applications," *Sensors*, vol. 18, no. 12, Art. no. 4212, 2018, doi: 10.3390/s18124212.

[10] A. Vena, E. Moradi, K. Koski, A. A. Babar, L. Sydänheimo, L. Ukkonen, and M. M. Tentzeris, "Design and realization of stretchable sewn chipless RFID tags and sensors for wearable applications," in *Proc. IEEE Int. Conf. RFID*, 2013, pp. 176–183.

[11] Q. Liu, J. Shen, J. Yin, H. Liu, and Y. Liu, "Compact 0.92/2.45-GHz dual-band directional circularly polarized microstrip antenna for handheld RFID reader applications," *IEEE Trans. Antennas Propag.*, vol. 63, no. 9, pp. 3849–3856, Sep. 2015, doi: 10.1109/TAP.2015.2452954.

[12] A. Romputtal and C. Phongcharoenpanich, "IoT-linked integrated NFC and dual band UHF/2.45 GHz RFID reader antenna scheme," *IEEE Access*, vol. 7, pp. 177832–177843, 2019, doi: 10.1109/ACCESS.2019.2958257.

[13] L. Chang, H. Wang, Z. Zhang, Y. Li, and Z. Feng, "Compact single-feed dual-mode antenna for active RFID tag application," *IEEE Trans. Antennas Propag.*, vol. 63, no. 11, pp. 5190–5194, Nov. 2015, doi: 10.1109/TAP.2015.2477417.

[14] G. Sen, A. Banerjee, M. Kumar, and S. Das, "An ultra-wideband monopole antenna with a gain enhanced performance using a novel split-ring meta-surface reflector," *Microw. Opt. Technol. Lett.*, vol. 59, no. 6, pp. 1296–1300, Jun. 2017, doi: 10.1002/mop.30527.

[15] N. A. Jan, S. H. Kiani, D. A. Sehrai, M. R. Anjum, A. Iqbal, M. Abdullah, and S. Kim, "Design of a compact monopole antenna for UWB applications," *Comput. Mater. Continua*, vol. 66, no. 1, pp. 35–44, 2021, doi: 10.32604/cmc.2020.012800.